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Applications of Dynamic Quick Response Codes From Farm to Customers

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Abstract

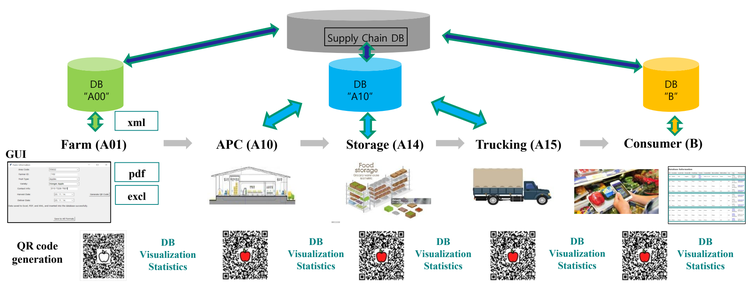
The agricultural product supply chain, spanning from farms to consumers, involves multiple interconnected stages. A lack of transparency across these stages often results in quality degradation, inefficient inventory management, and diminished consumer trust. Traditional static QR codes store only fixed information captured at the time of production, making it difficult to reflect new data generated during later stages of the supply chain. To address this limitation, digitalization of supply chain management requires systems capable of real-time data updating and sharing. Dynamic QR codes, connected to databases through redirect URLs, enable continuous information updates and transparent tracking of production, processing, and logistics throughout the entire product lifecycle. In this study, a real-time data management and traceability system based on dynamic QR codes was designed and implemented to cover the entire workflow from farms to consumers. QR codes generated at each stage were automatically synchronized with a centralized database, while a graphical user interface (GUI) allowed workers to intuitively enter and access information. Integrated QR code scanners facilitated automatic collection of key data such as location, environmental conditions, grading results, and processing times. As a result, all process information—from harvesting to consumption—was recorded in real time and made instantly accessible via QR codes. System application demonstrated practical benefits: farmers could review processing and distribution data to inform cultivation and management practices; workers could efficiently check production and storage information to optimize inventory and logistics management at agricultural processing centers (APCs); and consumers could scan product QR codes to verify production history, cultivation methods, quality grades, and distribution channels, thereby strengthening trust in food safety and product quality. This study highlights that dynamic QR codes serve not only as tools for delivering information but also as a digital backbone for modern agricultural supply chains. In particular, their ability to support real-time data updates and information sharing enhances traceability, streamlines inventory management, and promotes transparency across all stakeholders.

Key Words: Agricultural Products, Dynamic QR Code, Farm, Agricultural Products Processing Complex, Customer

Introduction

Utilizing Dynamic QR Code in Fresh Supply Chain

- **Supply Chain DB Architecture**
  - ❖ Database in Supply Chain
    - ➢ Product Producing & Processing Sector
    - ➢ Product Storage and Transportation (Logistics) Sector
    - ➢ Consumer Sector



1 Lo	2 AC	3 F...	4 FrT	5 Vt	6 Ct	7 HD	8 DD	9 Qt	10 Std
A0	55630	55	Apples	Hongro Apple	01012345678	2025-09-14	2025-09-14	100	2025-09-14
A10	55630	55	Apples	Hongro Apple	01012345678	2025-09-14	2025-09-14	100	2025-09-14
A11	55630	55	Apples	HongroApple	01012345678	2025-09-14	2025-09-14	100	2025-09-14
A12	55630	55	Apples	HongroApple	01012345678	2025-09-14	2025-09-14	100	2025-09-14
A13	55630	55	Apples	HongroApple	01012345678	2025-09-14	2025-09-14	100	2025-09-14
A14	55630	55	Apples	HongroApple	01012345678	2025-09-14	2025-09-14	100	2025-09-14
A15	55630	55	Apples	HongroApple	01012345678	2025-09-14	2025-09-14	100	2025-09-14

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15 APC_AD	16 APC_WD	17 APC_RT	18 APC_PT	19 APC_StD	20 AGrade	21 BGrade	22 CGrade	23 DefectRate	24 APC_OP	25 Tp	26 Hm
(NULL)	(NULL)	(NULL)	(NULL)	(NULL)	(NULL)	(NULL)	(NULL)	(NULL)	(NULL)	(NULL)	(NULL)
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2025-09-14 13:41:35	2025-09-14 13:46:17	2025-09-14 14:11:30	2025-09-14 14:20:28	2025-09-14 14:27:48	30	10	10	0	(NULL)	25.9	45
2025-09-14 13:41:35	2025-09-14 13:46:17	2025-09-14 14:11:30	2025-09-14 14:20:28	2025-09-14 14:27:48	30	10	10	0	2025-09-14 14:29:10	25.9	45

Object

- To explore the application of dynamic QR codes within Agricultural Products Processing Complexes (APCs).
- To design a system for real-time data tracking throughout key APC processes such as washing, sorting, and packaging.
- To create a web-based interface for consumers to access product traceability information by scanning a dynamic QR code.

Results and Discussion

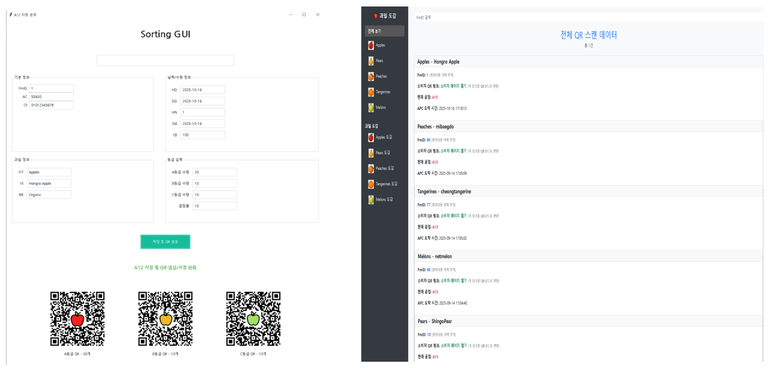
Detailed Scenarios and Web Visualization

- **Application of Dynamic QR Code at Farm Warehouse (A01)**
  - ❖ Generate and scan QR code at the Farm Warehouse (A01) for shipping to APC
    - ➢ Store data into the DB upon arrival at the farm warehouse: Farm ID, Date, Input time to the farm warehouse, Warehouse temperature and humidity, etc.

- **Application of Dynamic QR Code at Product-In Point**
  - ❖ Scan QR code generated from the Farm (A01) at Product-in Point(A10) in APC
    - ➢ QR code scanner acquires information on a box
    - ➢ Store data into the DB at APC: APC ID, date, Information generated form the farm, Product-in time, Quantity of Product, etc.

- **Application of Dynamic QR Code at Washing Point**
  - ❖ Scan QR code at Washing Point(A11) in APC
    - ➢ Store data into the DB at APC: APC ID, Date, Input time to Washing Point, Quantity of Product, etc.

- **Application of Dynamic QR Code at Sorting Point**
  - ❖ Scan QR code at Sorting Point(A12) in APC
    - ➢ Store data into the DB at APC: APC ID, Date, Input time to Sorting Point, Quantity of Product, Quality of Product etc.

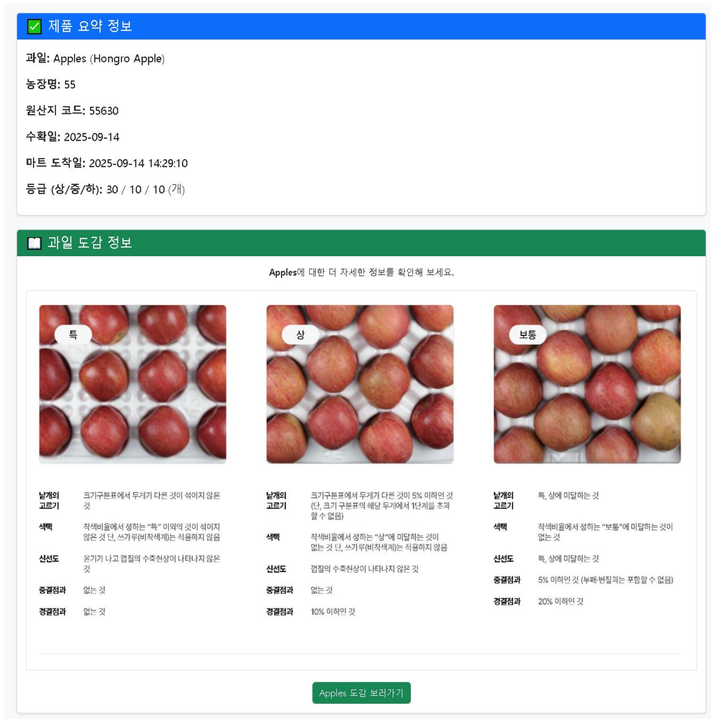


- **Application of Dynamic QR Code at Packing Point**
  - ❖ Scan QR code at Packing Point(A13) in APC
    - ➢ Store data into the DB at APC: APC ID, Date, Input time to Packing Point, Quantity of Product, Quality of Product etc.

- **Application of Dynamic QR Code at Storage Point**
  - ❖ Scan QR code at Storage Point(A14) in APC
    - ➢ Store data into the DB at APC: APC ID, Date, Input time to Storaging Point, Quantity of Product, Quality of Product etc.
    - ➢ If stored, the temperature and humidity of the storage facility can be monitored and saved to the DB.

- **Application of Dynamic QR Code at Shipping Point**
  - ❖ Scan QR code at Shipping Point(A15) in APC
    - ➢ Store data into the DB at APC: APC ID, Date, Input time to Out-Point, Quantity of Product, Quality of Product etc.

- **Application of Dynamic QR Code at Retail Point**
  - ❖ Consumers can access to information of the products in a retail shop
    - ➢ Website is organized for easy viewing by consumers when they scan the QR code.
    - ➢ Fruit type, Harvest location, Harvester, Market arrival date, Grade, etc.



Conclusions

This study has investigated a real-time data management and traceability system using dynamic QR codes, focusing on the process from the moment agricultural products are received at the Agricultural Processing Center (APC) until they reach the consumer. All information generated at each stage within the APC—including receiving, washing, sorting, packaging, and shipping—is recorded and synchronized in real time onto the dynamic QR code. Through this system, the APC can maximize the efficiency of its internal processes and perform precise, data-driven logistics management. Furthermore, consumers can enhance their trust in product quality by directly verifying the transparent processing history from the APC simply by scanning the QR code. In conclusion, dynamic QR codes have shown high potential as a key technology for enhancing supply chain transparency and efficiency by digitally integrating the complex internal processes of the APC and the distribution chain.

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